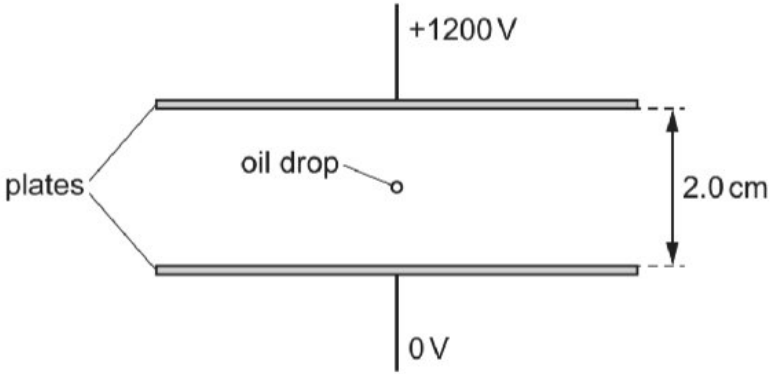
18 An oil drop of mass 2.6×10^{-15} kg and with a charge of -4.8×10^{-19} C is in a vacuum between two horizontal plates. The plates have a separation of 2.0 cm and a potential difference between them of 1200 V, as shown.



Which statement describes the motion of the oil drop?

A It is stationary.

B It has a downward acceleration of 9.7 m s^{-2}.

C It has an upward acceleration of 1.3 m s^{-2}.

D It has an upward acceleration of 11 m s^{-2}.